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\documentclass{article}

% Language setting
% Replace `english' with e.g. `spanish' to change the document language
\usepackage[english]{babel}
\usepackage{amssymb}

% Set page size and margins
% Replace `letterpaper' with `a4paper' for UK/EU standard size
\usepackage[letterpaper,top=2cm,bottom=2cm,left=3cm,right=3cm,marginparwidth=1.75cm]{geometry}

% Useful packages
\usepackage{amsmath}
%\usepackage{graphicx}
%\usepackage[colorlinks=true, allcolors=blue]{hyperref}


\title{A GENERALIZATION OF  $3x+1$  PROBLEM TO  $3x+4y+1$  }
\author{Robert Amirian, MSEE, University of Illinois, Urbana- Champaign \\\
Alan Amirian B.S. Data Science, UCSD}
\date{\today} % You can replace \today with a fixed date like "September 2025"
\begin{document}
\maketitle
\thispagestyle{empty} % Remove page number from title page

\newpage % Starts a new page after the title


\section{Abstract}
Keywords: I. Collatz, II. Generalized, III.  $3x+1$ \\\\
This paper seeks to generalize the infamous Collatz Conjecture which has haunted mathematicians for the last 100 years. Our research shows that there are infinite variations to the Collatz Conjecture, or the  $3x+1$  problem, in the form of  $3x+4y+1$  for all integer values of  $y$  positive and negative. The proposed Generalized Collatz Conjecture (GCC), or  $3x+4y+1$  problem, generalizes the Collatz Conjecture and shows that all GCC variations finish their cycle at the minimum value of  $1-2y$ , rather than just 1 as the Collatz Conjecture states. \\\\\
We believe this generalization opens the door to solving the Generalized Collatz Conjecture, and thus, Collatz Conjecture.


\section{Generalization - GCC}

Let  $x_{\{0\}}$  and  $y$  denote integers such that

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sequence for x_{n+1} : 5, 28, 11, 46, 20, 7, 34, 14, 4, -1, 10, 2, -2, -4, -5, -2, -4, -5, -2. The number of iterations is 14, ending with a minimum value of -5.\\

\section{Numerical Experiments}

Similar to the original Collatz problem, there is not yet a proof for this generalized sequence, and therefore to assess the accuracy of the generalized sequence, we resorted to numerical investigation.

The generalized conjecture has been verified numerically by testing the sequence for:\\

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\\(-400,000 \leq x_{0} \leq 400,000 \\)\\

\\(-400,000 \leq y \leq 400,000 \\)\\

for around 640 billion data points, as well as numerous other random combinations of x_{0} and y . All the numerical experiments conducted have proven the conjecture to be correct, as they all reach the values given by Eq. (2).\\

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